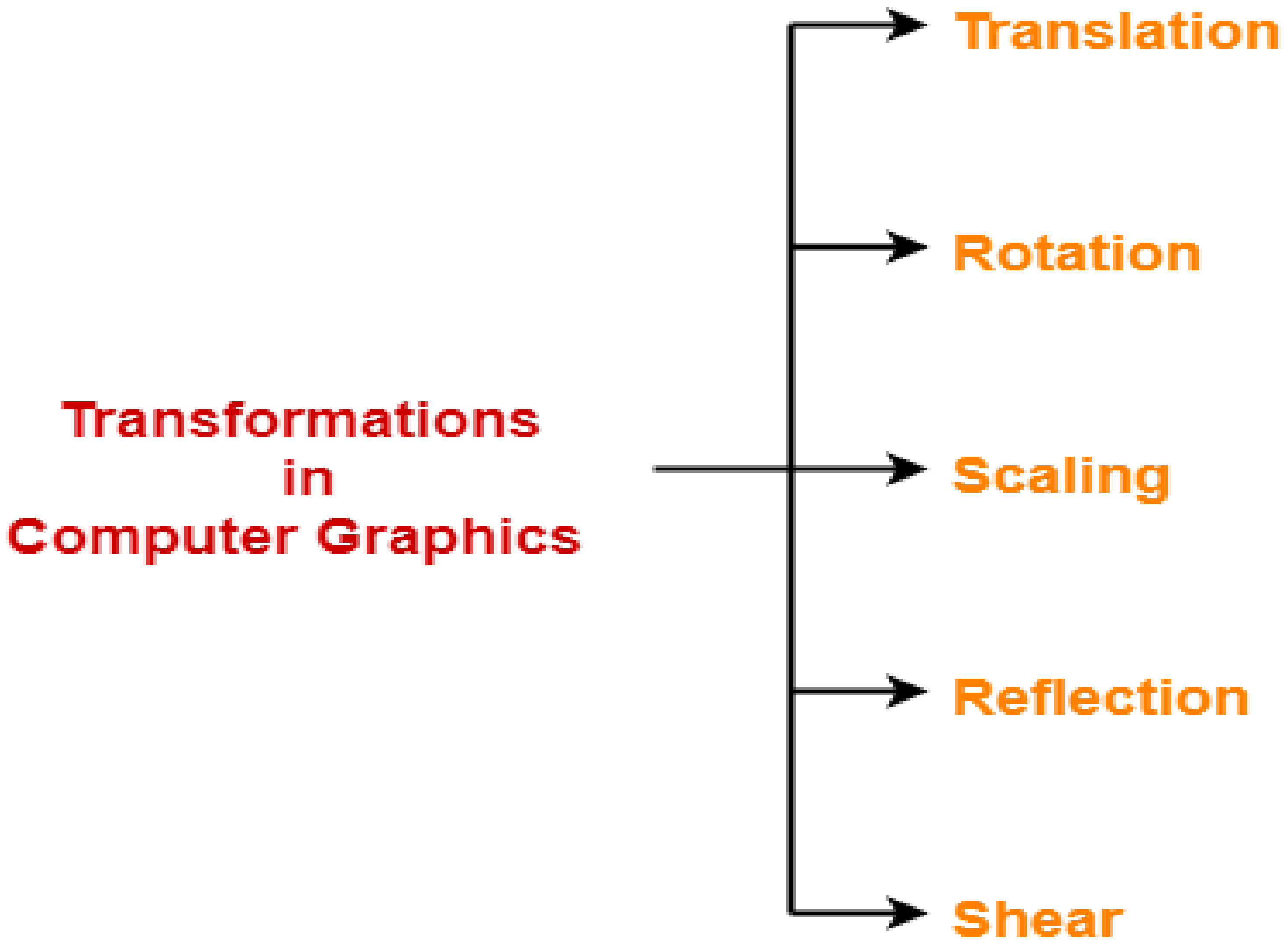
3D Transformation in Computer Graphics

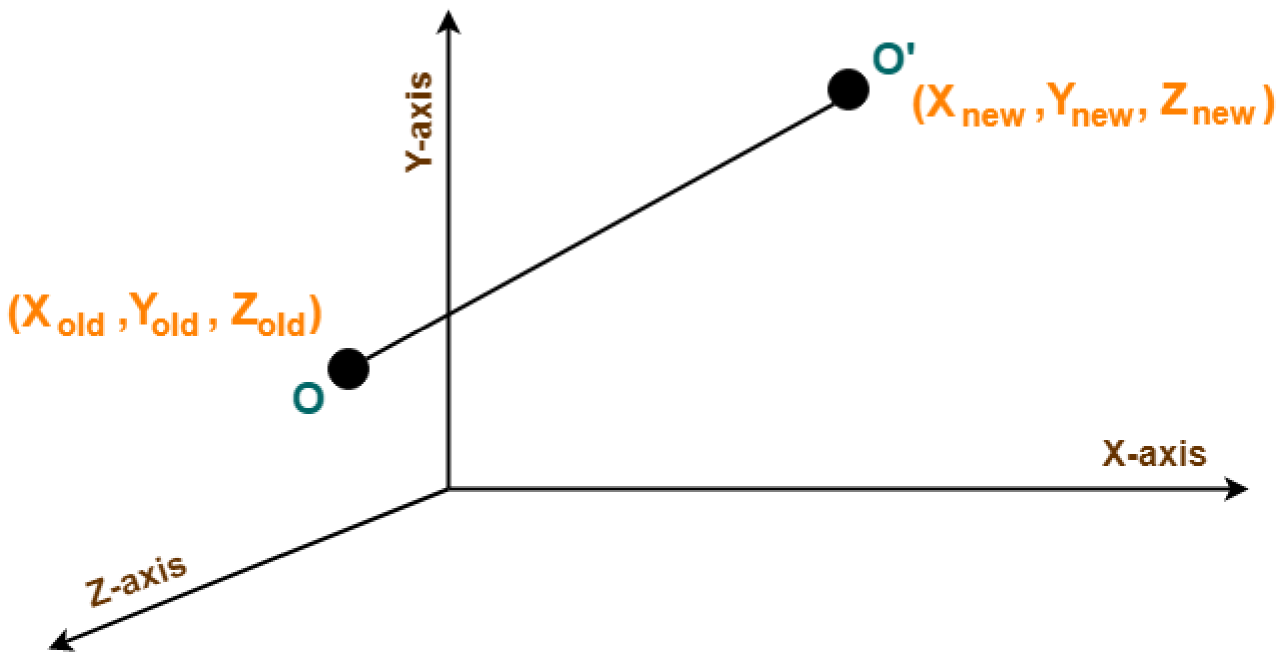


3D Transformation in Computer Graphics

- 3D Transformations take place in a three dimensional plane.
- 3D Transformations are important and a bit more complex than 2D Transformations.
- Transformations are helpful in changing the position, size, orientation, shape etc of the object.

3D Translation in Computer Graphics

- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- New coordinates of the object O after translation = $(X_{new}, Y_{new}, Z_{new})$
- Translation vector or Shift vector = (T_x, T_y, T_z)



- $X_{new} = X_{old} + T_x$ (This denotes translation towards X axis)
- $Y_{new} = Y_{old} + T_y$ (This denotes translation towards Y axis)
- $Z_{new} = Z_{old} + T_z$ (This denotes translation towards Z axis)

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Translation Matrix

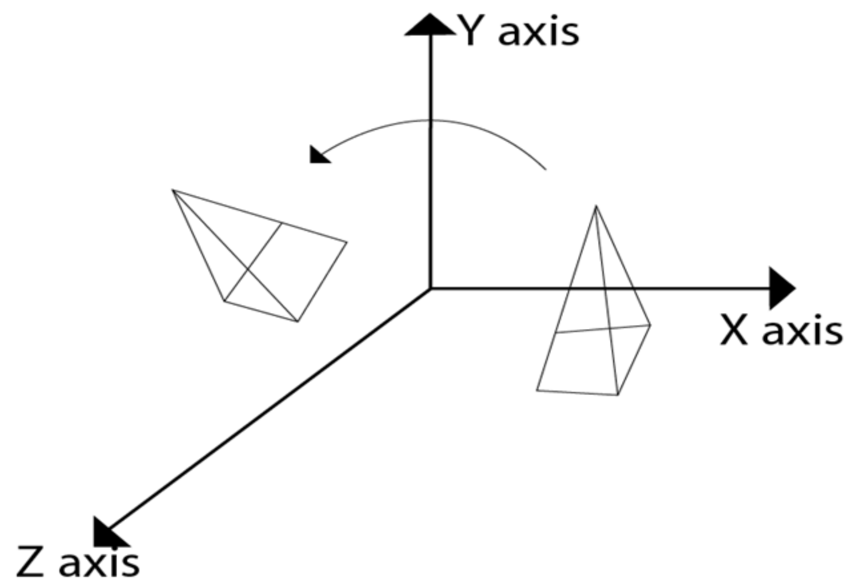
Given a 3D object with coordinate points A(0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0). Apply the translation with the distance 1 towards X axis, 1 towards Y axis and 2 towards Z axis and obtain the new coordinates of the object.

Thus, New coordinates of the object = A (1, 4, 3), B(4, 4, 4), C(4, 1, 2), D(1, 1, 2).

3D Rotation in Computer Graphics

In 3 dimensions, there are 3 possible types of rotation:

- X-axis Rotation
- Y-axis Rotation
- Z-axis Rotation



For X-Axis Rotation-

- $X_{new} = X_{old}$
- $Y_{new} = Y_{old} * \cos\theta - Z_{old} * \sin\theta$
- $Z_{new} = Y_{old} * \sin\theta + Z_{old} * \cos\theta$

For Y-Axis Rotation-

- $X_{new} = Z_{old} * \sin\theta + X_{old} * \cos\theta$
- $Y_{new} = Y_{old}$
- $Z_{new} = Z_{old} * \cos\theta - X_{old} * \sin\theta$

For Z-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{new} = X_{old} * \cos\theta - Y_{old} * \sin\theta$
- $Y_{new} = X_{old} * \sin\theta + Y_{old} * \cos\theta$
- $Z_{new} = Z_{old}$

Example

Given a homogeneous point (1, 2, 3). Apply rotation 90 degree towards X, Y and Z axis and find out the new coordinate points.

For X-Axis Rotation

- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} * \cos\theta - Z_{old} * \sin\theta = 2 * \cos 90° - 3 * \sin 90° =$
$2 * 0 - 3 * 1 = -3$
- $Z_{new} = Y_{old} * \sin\theta + Z_{old} * \cos\theta = 2 * \sin 90° + 3 * \cos 90° =$
$2 * 1 + 3 * 0 = 2$

Thus, New coordinates after rotation = (1, -3, 2).

For Y-Axis Rotation:

Let the new coordinates after rotation = (Xnew, Ynew, Znew).

Applying the rotation equations, we have-

- Xnew = Zold * sinθ + Xold * cosθ = 3 * sin90° + 1 x cos90° = 3 * 1 + 1 * 0 = 3
- Ynew = Yold = 2
- Znew = Zold * cosθ – Xold * sinθ =

For Z-Axis Rotation

Let the new coordinates after rotation = (Xnew, Ynew, Znew).

Applying the rotation equations, we have-

- Xnew = Xold * cosθ – Yold * sinθ =

1 * cos90° – 2 * sin90° = 1 * 0 – 2 * 1 = -2

- Ynew = Xold * sinθ + Yold * cosθ =

1 * sin90° + 2 * cos90° = 1 * 1 + 2 * 0 = 1

- Znew = Zold = 3

Thus, New coordinates after rotation = (-2, 1, 3)

3D Scaling in Computer Graphics

This scaling is achieved by using the following scaling equations-

- $X_{new} = X_{old} \times S_x$
- $Y_{new} = Y_{old} \times S_y$
- $Z_{new} = Z_{old} \times S_z$

3D Scaling in Computer Graphics

In Matrix form

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

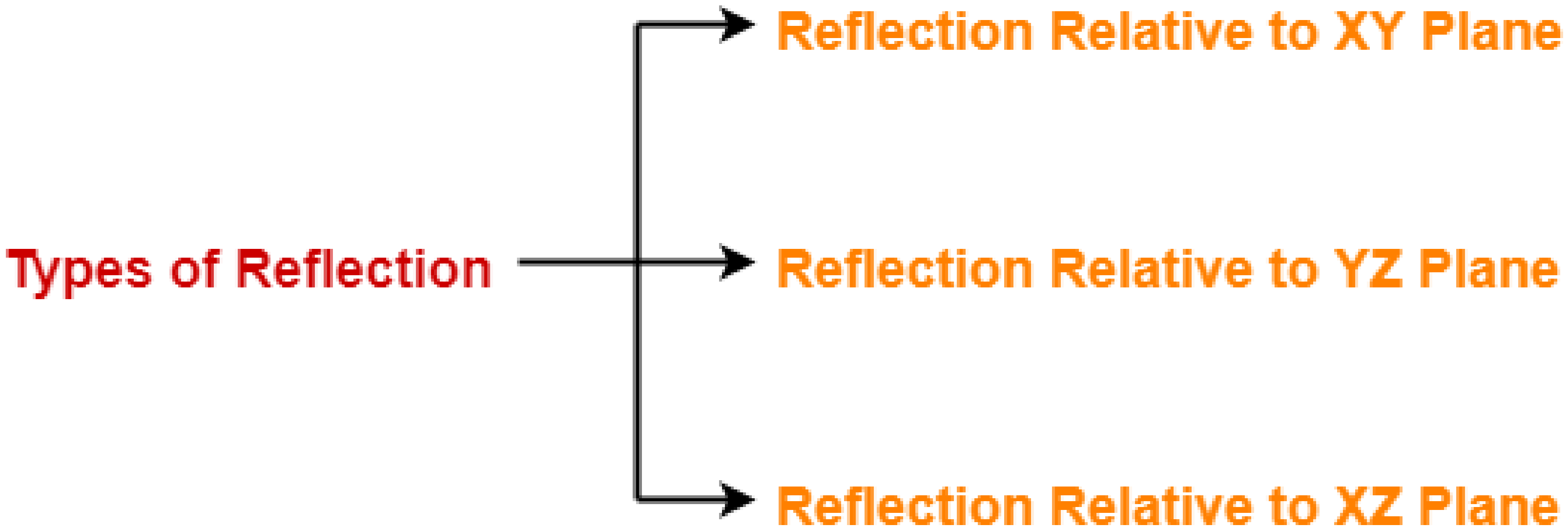
3D Scaling Matrix

Example

Given a 3D object with coordinate points A(0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0). Apply the scaling parameter 2 towards X axis, 3 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object.

- Thus, New coordinates of corner A after scaling = (0, 9, 9).
- Thus, New coordinates of corner B after scaling = (6, 9, 18).
- Thus, New coordinates of corner C after scaling = (6, 0, 3).
- Thus, New coordinates of corner D after scaling = (0, 0, 0).

3D Reflection in Computer Graphics



Reflection Relative to XY Plane:

This reflection is achieved by using the following reflection equations-

- $X_{new} = X_{old}$
- $Y_{new} = Y_{old}$
- $Z_{new} = -Z_{old}$

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Reflection Matrix

(Reflection Relative to XY plane)

Example

Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XY plane and find out the new coordinates of the object.

Thus, New coordinates of the triangle after reflection =
A (3, 4, -1), B(6, 4, -2), C(5, 6, -3).

Reflection Relative to YZ Plane:

This reflection is achieved by using the following reflection equations:

- Xnew = -Xold
- Ynew = Yold
- Znew = Zold

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Reflection Matrix

(Reflection Relative to YZ plane)

Reflection Relative to XZ Plane:

This reflection is achieved by using the following reflection equations-

- Xnew = Xold
- Ynew = -Yold
- Znew = Zold

Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XZ plane and find out the new coordinates of the object.

Thus, New coordinates of the triangle after reflection = A (3, -4, 1), B(6, -4, 2), C(5, -6, 3).

3D Shearing in Computer Graphics

1. Shearing in X direction
2. Shearing in Y direction
3. Shearing in Z direction

Shearing in X Axis-

Shearing in X axis is achieved by using the following shearing equations-

- $X_{new} = X_{old}$
- $Y_{new} = Y_{old} + Sh_y * X_{old}$
- $Z_{new} = Z_{old} + Sh_z * X_{old}$

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ Sh_y & 1 & 0 & 0 \\ Sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In X axis)

Shearing in Y Axis-

Shearing in Y axis is achieved by using the following shearing equations-

- $X_{new} = X_{old} + Sh_x * Y_{old}$
- $Y_{new} = Y_{old}$
- $Z_{new} = Z_{old} + Sh_z * Y_{old}$

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & Sh_x & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & Sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In Y axis)

Shearing in Z Axis

Shearing in Z axis is achieved by using the following shearing equations-

- Xnew = Xold + Shx * Zold
- Ynew = Yold + Shy * Zold
- Znew = Zold

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & Sh_x & 0 \\ 0 & 1 & Sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In Z axis)

Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Thus, New coordinates of the triangle after shearing in X axis = A (0, 0, 0), B(1, 3, 5), C(1, 3, 6).

Thus, New coordinates of the triangle after shearing in Y axis = A (0, 0, 0), B(3, 1, 5), C(3, 1, 6).

Thus, New coordinates of the triangle after shearing in Z axis = A (0, 0, 0), B(5, 5, 2), C(7, 7, 3).